Experiment 4

Aim: Implement K-Nearest Neighbors (KNN) and evaluate model performance

1.Dataset Source

Dataset used:**Stroke Prediction Dataset**

Source link:https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

	A	B	C	D	E	F	G	H	I	J	K	L
1	id	gender	age	hypertens	heart_dise	ever_marr	work_type	Residence	avg_gluco	bmi	smoking_	stroke
2	9046	Male	67	0	1	Yes	Private	Urban	228.69	36.6	formerly s	1
3	51676	Female	61	0	0	Yes	Self-empl	Rural	202.21	N/A	never smc	1
4	31112	Male	80	0	1	Yes	Private	Rural	105.92	32.5	never smc	1
5	60182	Female	49	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
6	1665	Female	79	1	0	Yes	Self-empl	Rural	174.12	24	never smc	1
7	56669	Male	81	0	0	Yes	Private	Urban	186.21	29	formerly s	1
8	53882	Male	74	1	1	Yes	Private	Rural	70.09	27.4	never smc	1
9	10434	Female	69	0	0	No	Private	Urban	94.39	22.8	never smc	1
10	27419	Female	59	0	0	Yes	Private	Rural	76.15	N/A	Unknown	1
11	60491	Female	78	0	0	Yes	Private	Urban	58.57	24.2	Unknown	1
12	12109	Female	81	1	0	Yes	Private	Rural	80.43	29.7	never smc	1
13	12095	Female	61	0	1	Yes	Govt_job	Rural	120.46	36.8	smokes	1
14	12175	Female	54	0	0	Yes	Private	Urban	104.51	27.3	smokes	1
15	8213	Male	78	0	1	Yes	Private	Urban	219.84	N/A	Unknown	1
16	5317	Female	79	0	1	Yes	Private	Urban	214.09	28.2	never smc	1
17	58202	Female	50	1	0	Yes	Self-empl	Rural	167.41	30.9	never smc	1
18	56112	Male	64	0	1	Yes	Private	Urban	191.61	37.5	smokes	1
19	34120	Male	75	1	0	Yes	Private	Urban	221.29	25.8	smokes	1

This dataset contains patient medical records used to predict **stroke risk**.

2. Dataset Description

The dataset contains **health information of patients** and whether they have experienced a stroke.

Dataset Characteristics

Property	Value
Total Records	5110
Features	11
Target Variable	stroke

Problem Type	Binary Classification

Features

Feature	Description
gender	Male/Female
age	Age of patient
hypertension	Whether patient has hypertension
heart_disease	Presence of heart disease
ever_married	Marital status
work_type	Type of employment
Residence_type	Urban or Rural
avg_glucose_level	Average glucose level
bmi	Body mass index
smoking_status	Smoking habit
stroke	Target variable

Target Variable

Value	Meaning
0	No stroke
1	Stroke occurred

Real-World Application

Hospitals and healthcare systems use such models to **predict stroke risk early**, helping doctors take preventive actions.

3. Mathematical Formulation of KNN

K-Nearest Neighbors is a **non-parametric supervised learning algorithm** used for classification and regression.

For classification, the algorithm assigns a class based on the **majority class of its nearest neighbors**.

Distance Calculation

Most commonly used distance metric:

Euclidean Distance

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Prediction Rule

The predicted class is determined by:

$$\hat{y} = \text{majority class among k nearest neighbors}$$

Where **k** represents the number of neighbors.

4. Algorithm Limitations

1. **Slow with large datasets**
   Because it calculates distance with every training sample.
2. **Sensitive to feature scaling**
   Features with larger values dominate distance calculations.
3. **Curse of dimensionality**
   Performance decreases when the number of features increases.
4. **Choice of k value**
   Too small → noisy model
   Too large → overly smooth decision boundary.

5. Methodology / Workflow

Steps followed in the experiment:

1. Dataset collection

```
uploaded = files.upload()

df = pd.read_csv(io.BytesIO(uploaded[list(uploaded.keys())[0]]))

df.head()
```

Choose Files healthcare-...ke-data.csv

healthcare-dataset-stroke-data.csv(text/csv) - 316971 bytes, last modified: 3/16/2026 - 100% done

```
Saving healthcare-dataset-stroke-data.csv to healthcare-dataset-stroke-data.csv
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

2. Data preprocessing
3. Encoding categorical features

```
le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])
```

4. Train-test split

5.
```
X = df.drop("stroke", axis=1)
y = df["stroke"]
```

6. Feature scaling

7.
```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

8. KNN model training

```
knn = KNeighborsClassifier(n_neighbors=5)

knn.fit(X_train, y_train)
```

```
KNeighborsClassifier
KNeighborsClassifier()
```

9. Prediction on test data

```
y_pred = knn.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("KNN Accuracy:", accuracy)
```

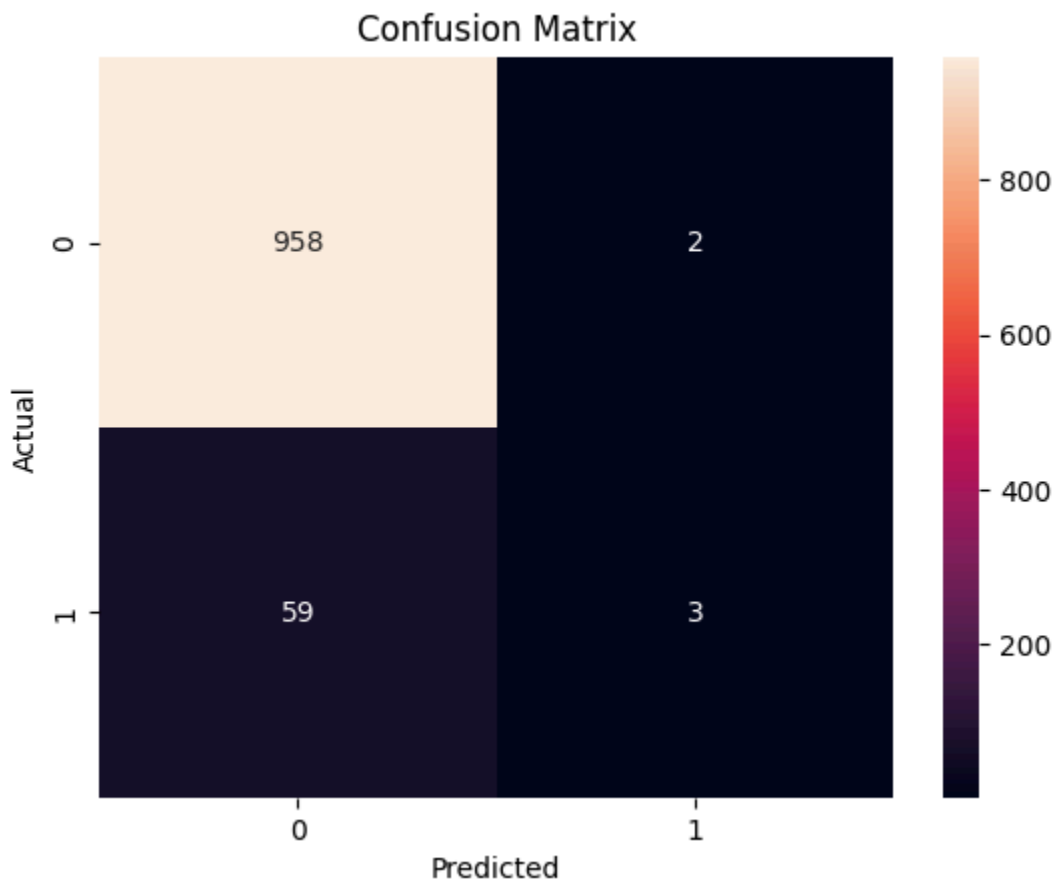
```
KNN Accuracy: 0.9403131115459883
```

10. Model evaluation

```
              precision    recall  f1-score   support

           0       0.94      1.00      0.97       960
           1       0.60      0.05      0.09        62

    accuracy                           0.94      1022
   macro avg       0.77      0.52      0.53      1022
weighted avg       0.92      0.94      0.92      1022
```



Workflow Diagram

Dataset

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Data Preprocessing

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Feature Scaling

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Train-Test Split
↓
KNN Model
↓
Prediction
↓
Performance Evaluation

6. Performance Analysis

Evaluation metrics used:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics measure how well the model predicts stroke risk.

7. Hyperparameter Tuning

Key KNN hyperparameters:

Parameter	Description
n_neighbors	Number of nearest neighbors
metric	Distance metric
weights	Uniform or distance-based

Tuning helps find the **optimal value of k** for best performance.